

Convergent retention of oxidative-defense orthogroups across independent extremotolerant origins: A comparative-proteomics screen in Deinococcaceae, Tardigrada, and Bdelloidea

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1 Introduction (skeleton)

Extremotolerant organisms have evolved radiation resistance independently at least three times: in the bacterial family Deinococcaceae, in tardigrades, and in bdelloid rotifers. Whether these origins converged on the same molecular machinery is unknown. We test whether orthogroups are convergently retained across resistant lineages while absent from radiation-sensitive controls.

2 Methods (pre-registered; amendments dated 2026-09-26)

Panel: 6 resistant proteomes (*Deinococcus radiodurans*, *D. geothermalis*, *D. deserti* [origin 1: Deinococcaceae]; *Ramazzottius varieornatus* [origin 2: Tardigrada]; *Adineta vaga* [origin 3: Bdelloidea]) vs 6 sensitive controls (*Thermus aquaticus*, *T. thermophilus*, *Hypsibius exemplaris*, *Caenorhabditis elegans*, *Drosophila melanogaster*, *Escherichia coli*). *Milnesium tardigradum* has no reference assembly; Tardigrada origin is single-proteome (power caveat carried throughout).

Orthogroup calling: MMseqs2 v17 sensitive clustering (min_seq_id 0.3, coverage 0.5, cov-mode 1), 154,665 proteins. Linclust used only as sensitivity; subset concordance (20k proteins): linclust pair purity 0.914 vs sensitive, retention 0.363, so all formal claims rest on sensitive clustering.

Formal test (locked 11:15 and 11:29 amendments, before outcome inspection): per orthogroup, one-sided Fisher exact on presence/absence (6 resistant vs 6 sensitive) and one-sided Mann-Whitney U on zero-filled copy numbers. The BH-FDR gate was replaced pre-outcome by 1000 label permutations with per-orthogroup empirical p , after W03's power analysis showed BH-FDR is structurally impossible at 6-vs-6 (minimum achievable Fisher $p = 1/\binom{12}{6} = 1.08 \times 10^{-3}$ vs BH rank-1 threshold $\approx 10^{-6}$). Gate: empirical $p < 0.05$, resistant-direction, replication in ≥ 2 of 3 independent resistant origins.

Candidate annotation: NCBI esummary + hmmscan (HMMER 3.3.2) vs Pfam-A (full-sequence $E < 10^{-10}$).

3 Results

Of 52,786 orthogroups tested, **12 pass all locked gates**: convergently present in Deinococcaceae and Bdelloidea, absent from all 6 sensitive controls, empirical $p = 0.023$ (1000 permutations). A further 53 orthogroups pass the copy-number Mann-Whitney arm.

Table 1: Convergent orthogroups passing all locked gates, with Pfam domain calls.

Representative	Pfam domains (E-value)	Interpretation
A. vaga UJR09360.1	AhpC-TSA (3.8e-35), Redoxin (8.8e-17)	Peroxiredoxin; peroxide detox
D. geothermalis WP_272975990.1	DSBA (7e-50)	Thiol-disulfide oxidoreductase
D. deserti WP_012692142.1	Flavin_Reduct (8.1e-26)	Flavin reductase; redox
D. deserti WP_012692117.1	Ank $\times 5$ (6e-24...)	Ankyrin scaffold; DDR interactions
D. geothermalis WP_414646549.1	PAD_porph (7.7e-122)	Porphyrin/heme binding
D. geothermalis WP_272975981.1	Acetyltransf_1/7	GNAT acetyltransferase
A. vaga UJR10666.1	HPPK (3.3e-29), Pterin_bind	Folate/pterin pathway
A. vaga UJR11615.1	DHH (2.8e-21), DHHA2 (2e-22)	Phosphoesterase; nucleotide damage
A. vaga UJR26281.1	HAD_2 (2.2e-25), Hydrolase	HAD-family hydrolase
A. vaga UJR28130.1	GDP_Man_Dehyd, NmrA	Redox-sensing dehydrogenase
A. vaga UJR36058.1	DUF4385 (2.6e-60)	Uncharacterized (novel space)
A. vaga UJR17759.1	Amidase (2.8e-61), Bac_rhamnosid6H	Cell-wall/envelope metabolism

The annotated candidates concentrate on the oxidative-defense axis: peroxiredoxin (AhpC-TSA), thioredoxin-family oxidoreductase (DSBA), and flavin reductase. Ionizing-radiation damage is dominated by reactive oxygen species; convergent retention of an oxidative-defense module across a bacterial and a metazoan origin, with complete absence from sensitive controls, is the screen’s central signal. The folate/pterin candidate (HPPK) is consistent with the known *Deinococcus* folate-photoprotection mechanism, and DHH/DHHA2 phosphoesterases act in nucleotide-damage processing.

3.1 Specificity layer on the extended panel (amendment 2026-09-26, 13:18 IST)

The panel was extended pre-data with four stress-tolerant but radiation-sensitive controls — *Pyrococcus furiosus*, *Sulfolobus solfataricus*, *Lactobacillus plantarum*, and *Saccharomyces cerevisiae* — chosen to separate radiation-specific retention from generic stress adaptation. All 168,186 proteins from the 15 proteomes were re-clustered de novo at the locked parameters (min. seq. id. 0.3, coverage 0.5, cov-mode 1), yielding 60,164 clusters. The locked gates were re-run identically on this v2 clustering, and each v1 candidate was then required to be absent from at least 3 of the 4 new controls to retain the label *radiation-specific*; otherwise it is re-labeled *broad-stress adaptation*.

All 12 v1 representative proteins mapped into v2 clusters. **10 of 12 candidates pass both the re-run enrichment gate (empirical $p = 0.013$ – 0.017 , 1000 permutations) and the specificity layer**, each with Deinococcaceae+Bdelloidea origins and at most one control-species member. Two honest negatives: (i) the v1 lead, the AhpC-TSA peroxiredoxin (UJR09360.1), is present in 3 of 4 stress-tolerant controls and is re-labeled broad-stress adaptation — biologically consistent, since peroxiredoxins are near-universal stress proteins; (ii) the HPPK/pterin candidate (UJR10666.1) lost its Deinococcaceae members in the larger panel (Bdelloid-only cluster, empirical $p = 0.449$) and no longer shows cross-kingdom convergence.

The oxidative-defense axis survives the specificity layer through the DSBA thiol-disulfide oxidoreductase and the flavin reductase candidates, now supported as radiation-specific rather than generic stress adaptations. The surviving set additionally includes the ankyrin scaffold, PAD-porphyrin binding protein, GNAT acetyltransferase, DHH/DHHA2 phosphoesterase, HAD hydrolase, GDP-mannose dehydrogenase, amidase, and the uncharacterized DUF4385 protein — the last representing genuinely novel candidate space.

3.2 Domain-level orthogonal confirmation (amendment 2026-09-26, 20:21 IST)

As an independent check, the panel was extended with five radiation-resistant proteomes never used in the screen: *Deinococcus radiophilus*, *D. proteolyticus*, *D. maricopensis*, *Adineta ricciae*, and *Rotaria magnacalcarata* (95,758 proteins; accessions and hashes in the ledger). For each radiation-specific candidate, its complete v1 Pfam domain set was searched against all five new proteomes (64 family HMMs; domain calls at $E \leq 10^{-5}$, fixed before candidate-level results). **9 of 10 candidates are confirmed:** their full domain architectures recur in at least one protein of at least two of the five new proteomes (eight at 5/5 support, one at 4/5). The five-domain amidase architecture (UJR17759.1) found no complete match and is labeled “domain-level support absent in extended panel” without removal from the candidate set (locked rule); the HPPK candidate that failed v2 enrichment is likewise unconfirmed, a consistent picture.

4 Caveats (honesty section)

(1) Tardigrada origin is single-proteome (Milnesium terminally unavailable); the gate’s “ ≥ 2 origins” was met by Deinococcaceae+Bdelloidea in every passing orthogroup - no Tardigrada-supported convergence is claimed. (2) Empirical $p = 0.023$ is the floor resolution at 1000 permutations for complete-separation configurations; it is not a continuous measure. (3) Empirical- p calibration assumes exchangeability under label permutation; family-level phylogenetic structure is only partially randomized away. (4) 4 esummary records failed uid resolution and await retry; their Pfam calls stand via hmmscan. (5) Candidates are computational; no experimental validation is claimed. (6) The v1 headline peroxiredoxin did not survive the specificity layer (Section 3.1); v1-era claims about AhpC-TSA as radiation-specific are superseded by the v2 labels.

5 Next experiments (falsifiable)

(1) Heterologous expression of the AhpC-TSA orthogroup in *E. coli*: gamma-survival assay. (2) Tardigrade proteome addition when a second assembly appears; re-run with identical locked gates. (3) Domain-level convergence test (hierarchy step 2 of the design amendment) as the orthogonal confirmation.